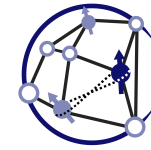


Post-Selected Quantum Metrology

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Quantum Info with Materials and Engineering

Outline

INTRO

METHOD

STRATEGY

RESULTS

ONGOING

1. **Post-Selection** — analytical bound on the post-selected QFI and its saturating probes
2. **DDrf** — hybrid driving, side-peak suppression, Gaussian pulse shaping
3. **Mattermost-Outline** — lab tooling update

Regardless of the quantity being sensed, quantum sensing requires an interaction between the system and the target. In our NV center system, we sense an external magnetic field, captured by the Hamiltonian $\mathcal{H}_B = \gamma B S_z$, where γ is the gyromagnetic ratio. Through this interaction term and the time evolution $e^{-i\mathcal{H}_B t}$, information about the B -field is imprinted onto the state: $|\psi\rangle \rightarrow e^{-i\mathcal{H}_B t}|\psi\rangle$.

The quantum Fisher information (QFI) quantifies how rapidly a quantum state — represented by a density matrix — changes with respect to the B -field. The faster the change, the greater the sensitivity:

$$F_Q = 2 \sum_{k,l} \frac{(\lambda_k - \lambda_l)^2}{\lambda_k + \lambda_l} |\langle k | \mathcal{H}_B | l \rangle|^2,$$

where $\rho = \sum_k \lambda_k |k\rangle \langle k|$.

We hypothesize that adding a post-selection step can be advantageous. Intuitively, the B -field information is encoded in the phase, and post-selection lets us discard the unnecessary part of the state into an ancillary qubit (or the other energy level). For a probe coupled to an ancilla,

$$\rho = p_0 |0\rangle\langle 0| \otimes \rho_0 + p_1 |1\rangle\langle 1| \otimes \rho_1 \longrightarrow \begin{cases} \rho_0 & \text{if the ancilla is measured in } |0\rangle, \text{ with probability } p_0, \\ \rho_1 & \text{if the ancilla is measured in } |1\rangle, \text{ with probability } p_1. \end{cases}$$

Last time I ran numerical simulations to identify which probe state and post-selection filter achieve the maximum QFI.

GH then shared a ChatGPT-generated report that gave useful insight into how to compute the bound.

Building on that, I reorganized the proof and analyzed the solution with Claude Opus 4.7.

Shared by GH — analytic two-qubit derivation.

The state evolves through four stages:

$$|\psi\rangle \xrightarrow{\text{Time evo}} |\psi_0\rangle = e^{-iH_B\tau} |\psi\rangle \xrightarrow{\text{Phase Damping}} \rho = \mathcal{D}(\rho_0) \xrightarrow{\text{Post-selection}} \frac{K\rho K^\dagger}{\text{Tr } K\rho K^\dagger}.$$

Without loss of generality, write the pure probe state in the computational basis:

$$|\psi\rangle = \sum_x a_x |x\rangle, \quad \sum_x |a_x|^2 = 1,$$

where x ranges over computational-basis bitstrings like $|00 \cdots 0\rangle$.

Time evolution. Under the detuned magnetic field,

$$|\psi_0\rangle = e^{-iH_B\tau} |\psi\rangle = \sum_x a_x e^{i\theta|x|} |x\rangle,$$

Post-Selection: Phase-Damping Channel

The density matrix before damping is

$$\rho_0 = \sum_{x,y} a_x \bar{a}_y e^{i\theta(|x|-|y|)} |x\rangle \langle y|.$$

After phase damping, off-diagonal elements decay by $\eta^{d(x,y)}$, where $d(x,y) = |x \oplus y|$ is the Hamming distance:

$$\begin{aligned} \rho_0 \rightarrow \rho &= \sum_{x,y} a_x \bar{a}_y e^{i\theta(|x|-|y|)} \eta^{d(x,y)} |x\rangle \langle y| \\ &= U(\theta) D_a C_N D_a^\dagger U^\dagger(\theta), \end{aligned}$$

with $D_a = \text{diag}(a_x)$ and $C_N = (I + \eta X)^{\otimes N}$.

Phase damping kills coherences across different Hamming weights but preserves populations $|a_x|^2$, so ρ has the eigenvalue structure of a *mixed* state on $\text{supp } \rho$.

Post-Selection: Filter Step

A θ -independent Kraus operator K (with $K^\dagger K \preceq I$) is applied to ρ , producing the success branch

$$\sigma = \frac{K\rho K^\dagger}{p_s}, \quad p_s = \text{Tr}[K\rho K^\dagger].$$

The conditional phase QFI

$$F_{\theta,Q}^{\text{ps}} = \text{Tr}(\sigma L_\sigma^2)$$

is the quantity we bound and optimize in the remaining slides.

Post-Selection: Quick Summary

For an N -qubit system, the phase QFI of the post-selected state obeys

$$F_{\theta,Q}^{\text{ps}} \leq \frac{N^2 \eta^2}{1 - \eta^2}, \quad \eta = e^{-\tau_d}, \quad \tau_d = (\tau/T_2)^p.$$

Strategy. Chain the relevant inequalities and analyze when each saturates. From the saturation conditions we can answer:

- Which probe states satisfy equality? What is the corresponding filter basis?
- Is there an entanglement-related gain?
- What is the post-selection success probability?
- Which probe state maximizes that success probability?

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- **Which probe states satisfy equality? What is the corresponding filter basis?**
Any $|\psi\rangle = \sum_x a_x |x\rangle$ with $a_x \neq 0$ for all x attains the maximum; the corresponding filter basis is derived below.
- **Is there an entanglement-related gain?**
No — the saturating class is dense and includes product states, so QFI alone shows no entanglement-related gain.
- **What is the post-selection success probability, and which state maximizes it?**
Among saturating probes, the state that maximizes the success probability is symmetric under both qubit exchange and bit flip.

The post-selected density matrix is

$$\sigma = \frac{K\rho K^\dagger}{p_s}, \quad p_s = \text{Tr } K\rho K^\dagger.$$

The QFI is obtained from the symmetric logarithmic derivative (SLD):

$$F_Q = \text{Tr } \sigma L^2, \quad \dot{\sigma} = \frac{1}{2}(\sigma L + L\sigma).$$

Computing the SLD directly is generally hard, so we work with another operator instead.

Post-Selection: Bounding via M

Introduce $M = \sigma^{-1/2} \dot{\sigma} \sigma^{-1/2}$. Substituting the SLD definition,

$$M = \frac{1}{2} (\sigma^{1/2} L \sigma^{-1/2} + \sigma^{-1/2} L \sigma^{1/2}).$$

The key inequality is $\boxed{\text{Tr } \sigma L^2 \leq \text{Tr } \sigma M^2}$.

In the eigenbasis $\sigma = \sum_k p_k |p_k\rangle \langle p_k|$ with $p_k \neq 0$,

$$L_{ab} = \frac{2\dot{\sigma}_{ab}}{p_a + p_b}, \quad M_{ab} = \frac{\dot{\sigma}_{ab}}{\sqrt{p_a p_b}}.$$

$$\text{Tr } \sigma L^2 = \sum_{a,b} \frac{2|\dot{\sigma}_{ab}|^2}{p_a + p_b}, \quad \text{Tr } \sigma M^2 = \sum_{a,b} \frac{|\dot{\sigma}_{ab}|^2 (p_a + p_b)}{2p_a p_b}.$$

Term by term, $m_{ab}/l_{ab} = (p_a + p_b)^2 / (4p_a p_b) \geq 1$, with **saturation** when $p_a = p_b$.

Post-Selection: Lifting to ρ

Here, σ lives in the subspace of ρ selected by K . Accordingly, M lives in that same subspace, and its operator norm is bounded by that of M_ρ :

$$\mathrm{Tr} \sigma M^2 \leq \|M_\rho\|_\infty^2.$$

On the other hand,

$$M_\rho = \rho^{-1/2} \dot{\rho} \rho^{-1/2}, \quad \dot{\rho} = -i[H_B, \rho] = \sum_i -\frac{i}{2} [Z_i, \rho].$$

Writing $M_\rho = \sum_i M_i$ with $M_i = \rho^{-1/2} \left(-\frac{i}{2} [Z_i, \rho]\right) \rho^{-1/2}$,

$$\|M_\rho\|_\infty \leq \sum_i \|M_i\|_\infty.$$

Here, $C_N = C^{\otimes N}$ and $[H_B, D_a] = 0$ force equality for any full-support probe.

Putting it together,

$$F_Q = \text{Tr } \sigma L^2 \leq \text{Tr } \sigma M_\sigma^2 \leq \|M_\rho\|_\infty^2 \leq \left(\sum_i \|M_i\|_\infty \right)^2.$$

When every step saturates simultaneously,

$$F_Q \leq N^2 \|M_i\|_\infty^2.$$

Three saturation conditions:

1. σ is maximally mixed on the filter subspace.
2. The optimal filter is chosen.
3. (Automatically satisfied here — see the previous slide.)

Next, we evaluate $\|M_i\|_\infty$.

To extract the eigenvalues and eigenvectors of M , it helps to recast the problem as a generalized eigenvalue problem:

$$\begin{aligned} M|v\rangle &= \lambda|v\rangle \\ \rho^{-1/2} \dot{\rho}_j \rho^{-1/2} |v\rangle &= \lambda|v\rangle \\ \dot{\rho}_j (\rho^{-1/2} |v\rangle) &= \lambda \rho (\rho^{-1/2} |v\rangle) \\ \dot{\rho}_j |w\rangle &= \lambda \rho |w\rangle, \end{aligned}$$

with $|w\rangle = \rho^{-1/2} |v\rangle$. The last line is the GEP we solve.

Post-Selection: Generalized Eigenvalue Problem

The standard eigenvalue problem is $A\mathbf{x} = \lambda\mathbf{x}$, with eigenvalues from $\det(A - \lambda I) = 0$

.

The **generalized eigenvalue problem (GEP)** extends this to $A\mathbf{x} = \lambda B\mathbf{x}$. This should be familiar from, e.g., coupled mechanical systems: the Lagrangian gives

$$M\ddot{\mathbf{x}} + K\mathbf{x} = 0.$$

A harmonic ansatz $\mathbf{x} = \mathbf{x}_0 e^{i\omega t}$ then yields

$$K\mathbf{x} = \omega^2 M\mathbf{x},$$

a GEP whose generalized eigenvalues are the squared normal-mode frequencies.

The relevant block-form is

$$\dot{\rho}_j = \begin{pmatrix} 0 & -i\eta B_j \\ i\eta B_j^\dagger & 0 \end{pmatrix},$$

so the GEP reduces to

$$\begin{pmatrix} 0 & -i\eta B_j \\ i\eta B_j^\dagger & 0 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \lambda \begin{pmatrix} A_j & \eta B_j \\ \eta B_j^\dagger & D_j \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}.$$

Post-Selection: Schur Decomposition

Applying a Schur-style decomposition with $B_j = A_j^{1/2} W_j D_j^{1/2}$,

$$\begin{pmatrix} A_j^{1/2} & 0 \\ 0 & D_j^{1/2} \end{pmatrix} \begin{pmatrix} 0 & -i\eta W_j \\ i\eta W_j^\dagger & 0 \end{pmatrix} \begin{pmatrix} A_j^{1/2} & 0 \\ 0 & D_j^{1/2} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \\ = \lambda \begin{pmatrix} A_j^{1/2} & 0 \\ 0 & D_j^{1/2} \end{pmatrix} \begin{pmatrix} I & \eta W_j \\ \eta W_j^\dagger & I \end{pmatrix} \begin{pmatrix} A_j^{1/2} & 0 \\ 0 & D_j^{1/2} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}.$$

Stripping the outer factors,

$$\begin{pmatrix} 0 & -i\eta W_j \\ i\eta W_j^\dagger & 0 \end{pmatrix} \begin{pmatrix} \tilde{w}_1 \\ \tilde{w}_2 \end{pmatrix} = \lambda \begin{pmatrix} I & \eta W_j \\ \eta W_j^\dagger & I \end{pmatrix} \begin{pmatrix} \tilde{w}_1 \\ \tilde{w}_2 \end{pmatrix}.$$

Post-Selection: Reducing via SVD

Take the SVD $W_j = U\Sigma_j V^\dagger$ with $\Sigma_j = \text{diag}(s_{jk})$:

$$\begin{pmatrix} 0 & -i\eta s_{jk} \\ i\eta s_{jk} & 0 \end{pmatrix} \begin{pmatrix} \tilde{u}_{jk1} \\ \tilde{u}_{jk2} \end{pmatrix} = \lambda_{jk} \begin{pmatrix} 1 & \eta s_{jk} \\ \eta s_{jk} & 1 \end{pmatrix} \begin{pmatrix} \tilde{u}_{jk1} \\ \tilde{u}_{jk2} \end{pmatrix}.$$

The determinant condition

$$\det \begin{pmatrix} -\lambda & -\eta s_{jk}(\lambda + i) \\ \eta s_{jk}(-\lambda + i) & -\lambda \end{pmatrix} = 0$$

gives

$$\lambda_{jk} = \pm \sqrt{\frac{\eta^2 s_{jk}^2}{1 - \eta^2 s_{jk}^2}}.$$

Post-Selection: Single-Qubit Bound

Since $0 \leq s_{jk} \leq 1$, the magnitude of λ_{jk} is maximized at $s_{jk} = 1$, giving

$$\|M_j\|_\infty \leq \frac{\eta}{\sqrt{1 - \eta^2}}.$$

This is one more inequality, which must hold for every qubit. Generically $s_{jk} < 1$ for mixed states; however, for any initial state $|\psi_0\rangle = \sum_x a_x |x\rangle$ with $a_x \neq 0$ for all x , phase damping still lets us attain $s_{jk} = 1$.

At joint saturation,

$$\|M\|_\infty = \frac{N\eta}{\sqrt{1 - \eta^2}}, \quad F_Q \leq \|M\|_\infty^2 = \frac{N^2\eta^2}{1 - \eta^2}.$$

Post-Selection: Saturating Probe States

Every pure state $|\psi_0\rangle = \sum_x a_x |x\rangle$ with $a_x \neq 0$ for **all** computational-basis strings x saturates the global bound

$$F_{\theta,Q}^{\text{ps}} = \frac{N^2 \eta^2}{1 - \eta^2}.$$

The saturating class is **open** (the condition $|a_x| > 0$ for all x is open) and **dense** (full-support probes are generic) in the projective Hilbert space $\mathbb{CP}^{2^N-1}$.

As shown above, the phase-damped state factorizes as $\rho = D_a C_N D_a^\dagger$ with $C_N = (I + \eta X)^{\otimes N}$. D_a is invertible iff $a_x \neq 0$ for all x . Return to the GEP:

$$\dot{\rho}_j |w\rangle = \lambda \rho |w\rangle.$$

Then

$$\begin{aligned} -i[H_B, D_a C_N D_a^\dagger] |w\rangle &= \lambda D_a C_N D_a^\dagger |w\rangle \\ -iD_a [H_B, C_N] D_a^\dagger |w\rangle &= \lambda D_a C_N D_a^\dagger |w\rangle, \end{aligned}$$

since H_B and D_a commute. Letting $|\epsilon\rangle = D_a^\dagger |w\rangle$, we obtain another GEP:

$$-i[H_B, C_N] |\epsilon\rangle = \lambda C_N |\epsilon\rangle.$$

Post-Selection: Single-Qubit GEP

With $C_N = (I + \eta X)^{\otimes N} = C^{\otimes N}$ and $H_B = \frac{1}{2} \sum_i Z_i$,

$$-\frac{i}{2} \sum_i [Z_i, C^{\otimes N}] |\epsilon\rangle = \lambda C^{\otimes N} |\epsilon\rangle.$$

For the product ansatz $|\epsilon\rangle = \bigotimes_i |\epsilon_i\rangle$, each factor satisfies a single-qubit GEP:

$$-\frac{i}{2} [Z, C] |\epsilon_i\rangle = \lambda C |\epsilon_i\rangle.$$

Since $-\frac{i}{2} [Z, C] = \eta Y$, this reduces to

$$\begin{pmatrix} -\lambda & -\eta(\lambda + i) \\ -\eta(\lambda - i) & -\lambda \end{pmatrix} |\epsilon_i\rangle = 0,$$

yielding $\lambda_{\pm} = \pm\eta/\sqrt{1-\eta^2}$ and $|\epsilon_i^{\pm}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{\pm i\alpha}|1\rangle)$, where $\cos \alpha = -\eta$,
 $\sin \alpha = \sqrt{1-\eta^2}$.

Post-Selection: Filter Vectors $|w^\pm\rangle$

Returning to $|w\rangle$:

$$|w^\pm\rangle = (D_a^{-1})^\dagger |\epsilon^\pm\rangle,$$

so

$$\dot{\rho} |w^\pm\rangle = \pm \frac{N\eta}{\sqrt{1-\eta^2}} \rho |w^\pm\rangle.$$

Therefore $M = \rho^{-1/2} \dot{\rho} \rho^{-1/2}$ contains the eigenvalues $\pm N\eta/\sqrt{1-\eta^2}$, giving $\|M\|_\infty = N\eta/\sqrt{1-\eta^2}$ and

$$F_{\theta,Q}^{\text{ps}} = \frac{N^2\eta^2}{1-\eta^2},$$

with filter vectors $|w^\pm\rangle$.

Post-Selection: Filter Basis $|\Psi^\pm\rangle$

Since $|w\rangle = \rho^{-1/2}|v\rangle$, the filter basis (denoted $|\Psi^\pm\rangle$) is

$$|\Psi^\pm\rangle = |v^\pm\rangle = \frac{\rho^{1/2}|w^\pm\rangle}{(1 - \eta^2)^{N/2}}.$$

The two basis vectors are mutually orthogonal:

$$\langle\Psi^+|\Psi^-\rangle = \frac{\langle w^+|\rho|w^-\rangle}{(1 - \eta^2)^N} = 0,$$

inherited from ρ -biorthogonality (the C -biorthogonality lifted to N qubits).

Post-Selection: Matched Kraus Operator

The filter is a rank-2 projector onto $\text{span}\{|\Psi^+\rangle, |\Psi^-\rangle\}$:

$$P_{\pm} = |\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-|.$$

The matched Kraus operator is

$$K = \sqrt{\frac{p_s}{2}} P_{\pm} \rho^{-1/2}.$$

Applying it to ρ :

$$K\rho K^{\dagger} = \frac{p_s}{2} P_{\pm} \rho^{-1/2} \rho \rho^{-1/2} P_{\pm} = \frac{p_s}{2} P_{\pm}^2 = \frac{p_s}{2} P_{\pm}.$$

Post-Selection: Accepted State

The accepted state is

$$\sigma = \frac{K\rho K^\dagger}{p_s} = \frac{P_\pm}{2} = \frac{|\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-|}{2}.$$

That is, the accepted state is the *maximally mixed state on the filter subspace* — exactly the structural requirement for QFI saturation.

This form looks unfamiliar, so we re-express the filter as

$$K = |\Psi^+\rangle\langle w^+| + |\Psi^-\rangle\langle w^-|.$$

It can be implemented as $K = \sqrt{1-\gamma} |00\cdots 0\rangle\langle 00\cdots 0| + |00\cdots 0\rangle\langle 10\cdots 0|$ by sandwiching with appropriate unitaries before and after the post-selection step, **setting one post-selection strength to zero and the others to one.**

The POVM constraint is $K^\dagger K \preceq I$. Explicitly,

$$K^\dagger K = |w^+\rangle\langle w^+| + |w^-\rangle\langle w^-| = (D_a^{-1})^\dagger \Pi D_a^{-1},$$

where

$$\Pi = |\epsilon^+\rangle^{\otimes N} \langle \epsilon^+|^{\otimes N} + |\epsilon^-\rangle^{\otimes N} \langle \epsilon^-|^{\otimes N}.$$

Its component form is

$$\Pi_{xy} = \frac{1}{2^{N-1}} \cos(\alpha(|x| - |y|)),$$

so $(K^\dagger K)_{xy} = \Pi_{xy} / (\bar{a}_x a_y)$.

Post-Selection: Success Probability — Setup

The condition $\Pi|u\rangle = \mu D_p|u\rangle$ is equivalent to $D_p^{-1/2}\Pi D_p^{-1/2}|w\rangle = \mu|w\rangle$ with $|w\rangle = D_p^{1/2}|u\rangle$, so we want the eigenvalues of the symmetric operator $D_p^{-1/2}\Pi D_p^{-1/2}$.

In the 2D image of Π (basis $\{|v^+\rangle^{\otimes N}, |v^-\rangle^{\otimes N}\}$), Π acts as the rank-2 identity.

The Gram matrix of $\{D_p^{-1/2}|v^\pm\rangle^{\otimes N}\}$ is

$$G = \begin{pmatrix} \hat{S} & \hat{T} \\ \overline{\hat{T}} & \hat{S} \end{pmatrix},$$

and the eigenvalues of $D_p^{-1/2}\Pi D_p^{-1/2}$ in this basis are those of G , namely $\hat{S} \pm |\hat{T}|$.

So $\lambda_{\max}(\Xi) = \hat{S} + |\hat{T}| = (S + |T|)/2^N$, giving

$$p_s^* = \frac{2(1 - c^2)^N}{\lambda_{\max}(\Xi)} = \frac{2^{N+1}(1 - c^2)^N}{S + |T|}.$$

Post-Selection: Optimal Probe — Symmetries

The optimization is over all probe magnitudes $\{|a_x|^2\}_x$ with $\sum_x |a_x|^2 = 1$. The objective $S + |T|$ depends on the magnitudes through

$$S = \sum_x \frac{1}{|a_x|^2}, \quad T = \sum_x \frac{e^{-2i\beta|x|}}{|a_x|^2}, \quad \beta = \pi - \arccos c.$$

The objective is **invariant under permutations of the N qubits**, since both S and $|T|$ depend on $|a_x|^2$ only through the Hamming weight $|x|$. It is also **invariant under bit-flip $x \rightarrow \bar{x}$** : $e^{-2i\beta|\bar{x}|} = e^{-2i\beta(N-|x|)} = e^{-2i\beta N} e^{2i\beta|x|}$ merely conjugates T — a global phase invisible to $|T|$.

By a standard symmetrization argument (averaging over the symmetry group does not increase $S + |T|$, by convexity), **the optimum lies in the permutation- and bit-flip-symmetric subclass.**

Post-Selection: Optimal Probe — Dicke Parametrization

In the symmetric subclass, $|a_x|^2$ depends only on $|x| = k$. Parametrize by Dicke populations p_k for $k = 0, 1, \dots, N$:

$$|a_x|^2 = \frac{p_k}{\binom{N}{k}} \text{ for } |x| = k, \quad \sum_k p_k = 1, \quad p_k = p_{N-k}.$$

The corresponding probe, in the Dicke basis $|D_k^N\rangle = \binom{N}{k}^{-1/2} \sum_{|x|=k} |x\rangle$, is

$$|\psi_0\rangle = \sum_{k=0}^N q_k |D_k^N\rangle, \quad q_k = \sqrt{p_k} \text{ (up to phase).}$$

This collapses the optimization from 2^N variables to $\lfloor N/2 \rfloor + 1$.

DDrf = selective, phase-controlled RF driving of nuclear spins, interleaved with dynamical decoupling on the electron spin.

If the Hamiltonian is **block-diagonal** in the electron basis,

$$H = |0\rangle\langle 0| \otimes H_0 + |1\rangle\langle 1| \otimes H_1,$$

then sandwiching the evolution with an electron π -pulse swaps the two branches:

$$|+\rangle|0\rangle \xrightarrow{t_1, \pi, t_2} \frac{1}{\sqrt{2}} \left(|0\rangle \otimes \underbrace{e^{-iH_0 t_2} e^{-iH_1 t_1}}_{U_0} |0\rangle + |1\rangle \otimes \underbrace{e^{-iH_1 t_2} e^{-iH_0 t_1}}_{U_1} |0\rangle \right).$$

Generically $U_0 \neq U_1$ — a **conditional gate**.

For NV- ^{13}C with RF driving:

$$\begin{aligned} H &= |0\rangle\langle 0| \otimes H_0 + |-1\rangle\langle -1| \otimes H_1, \\ H_0 &= \omega_0 I_z + 2\Omega_{\text{RF}} \cos(\omega_{\text{RF}}t + \phi) I_x, \\ H_1 &= \omega_1 \tilde{I}_z + 2\Omega_{\text{RF}} \cos \beta \cos(\omega_{\text{RF}}t + \phi) \tilde{I}_x + 2\Omega_{\text{RF}} \sin \beta \cos(\omega_{\text{RF}}t + \phi) \tilde{I}_z. \end{aligned}$$

The $|1\rangle$ branch has a **tilted** nuclear quantization axis (angle β , with $\sin \beta = A_{\perp}/\omega_1$). This small tilt is what enables hyperfine-mediated control of nuclei with vanishing $A_{\perp}$.

In two electron-conditioned rotating frames

$$R_s(t) = e^{i\omega_{\text{RF}}t \tilde{I}_z^{(s)}},$$

$$H'_0 = (\omega_0 - \omega_{\text{RF}}) I_z + \Omega_{\text{RF}}(\cos \phi I_x + \sin \phi I_y),$$

$$H'_1 = (\omega_1 - \omega_{\text{RF}}) \tilde{I}_z + \Omega_{\text{RF}} \cos \beta (\cos \phi \tilde{I}_x + \sin \phi \tilde{I}_y).$$

At resonance ($\omega_{\text{RF}} = \omega_1$), with $\omega_0 - \omega_{\text{RF}} \gg \Omega_{\text{RF}}$ and $\beta \rightarrow 0$, H'_1 becomes a pure transverse drive while H'_0 is a pure z -rotation.

Each MW π -pulse acts as an **instantaneous frame swap** $\Lambda_{s,\bar{s}}(t) \equiv R_s(t)R_{\bar{s}}(t)^\dagger$ between the two rotating frames. The full unitary $U = \sum_s |s\rangle\langle s| \otimes U_s$ is a chain of H'_s -segments separated by frame swaps:

$$U_s = R_s(4N\tau)^\dagger \cdot e^{-iH'_s\tau} \cdot \Lambda_{s,\bar{s}}((2N-1)\tau) \cdot e^{-iH'_{\bar{s}}2\tau} \cdot \Lambda_{\bar{s},s}((2N-3)\tau) \cdot e^{-iH'_s\tau} \dots \\ \dots e^{-iH'_s\tau} \cdot \Lambda_{s,\bar{s}}(3\tau) \cdot e^{-iH'_{\bar{s}}2\tau} \cdot \Lambda_{\bar{s},s}(\tau) \cdot e^{-iH'_s\tau} \cdot R_s(0).$$

This is exact under the assumption of negligible MW pulse duration, and **much faster than direct Schrödinger integration** when Ω_{RF} is constant: each $e^{-iH'_s\tau}$ is a single matrix exponential of a time-independent Hamiltonian.

Sequence: $\pi/2 \rightarrow \text{DDrf}(N, \tau) \rightarrow \pi/2_\phi$, projecting onto $|+\rangle$ ($\phi = \pi/2$).

For N nuclear spins,

$$P_x = \frac{1}{2} + \frac{1}{2^{N+1}} \Re \text{Tr } U_0 U_1^\dagger,$$

$$\text{Tr } U_0 U_1^\dagger = \prod_i \text{Tr } U_0^i U_1^{i\dagger}.$$

Peaks appear at $\omega_{\text{RF}} = \omega_1$.

The DDrf sequence is built from identical 4τ cells. Each cell is itself block-diagonal:

$$V^{(k)} = |0\rangle\langle 0| \otimes V_0^{(k)} + |1\rangle\langle 1| \otimes V_1^{(k)}, \quad U = \sum_{s \in \{0,1\}} |s\rangle\langle s| \otimes \prod_{k=1}^{N/2} V_s^{(k)}.$$

A z -rotation can be commuted through a transverse rotation by shifting its azimuthal angle:

$$e^{i\alpha I_z} (\cos \phi I_x + \sin \phi I_y) e^{-i\alpha I_z} = \cos(\phi - \alpha) I_x + \sin(\phi - \alpha) I_y.$$

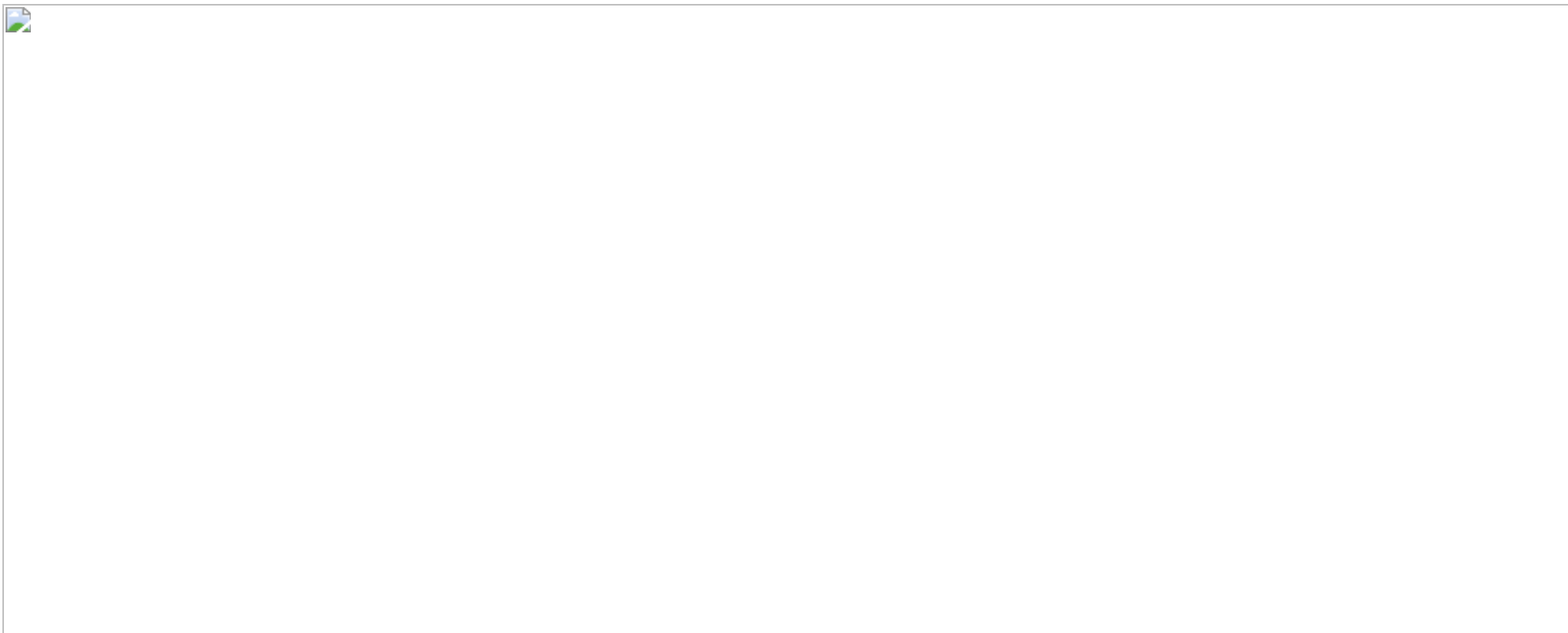
Applied per cell (Taminiau limit, $\beta = 0$, $\omega_{\text{RF}} = \omega_1$):

$$V_0^{(k)} = e^{-iH'_0\tau} e^{-iH'_1 2\tau} e^{-iH'_0\tau} = e^{-i 2\delta_0\tau I_z} e^{-i 2\Omega\tau \hat{\phi}'_k \cdot \vec{I}}.$$

Each cell splits cleanly into a z -piece plus a transverse rotation. Choosing ϕ_k to align successive transverse axes makes the product over k contract into a single conditional rotation.

In the Taminiau limit, $U_s = R_z(N(\omega_L - \omega_1)\tau) \cdot R_\phi(\pm N\Omega_{\text{RF}}\tau)$.

Observation. When $N\Omega_{\text{RF}}\tau = 2\pi$, $U_0 = U_1$ — the gate becomes **unconditional**, so a flat spectroscopy signal is expected. In practice, however:



Restoring finite detuning $\delta_1 = \omega_1 - \omega_{\text{RF}}$ (with $\beta = 0$ for clarity):

$$H'_1 = \delta_1 I_z + \Omega (\cos \phi I_x + \sin \phi I_y) = \Omega_{\text{eff}} \hat{n}(\phi) \cdot \vec{I},$$

$$\Omega_{\text{eff}} = \sqrt{\Omega^2 + \delta_1^2}, \quad \sin \gamma = \frac{\delta_1}{\Omega_{\text{eff}}}, \quad \hat{n}(\phi) = (\cos \gamma \cos \phi, \cos \gamma \sin \phi, \sin \gamma).$$

Conjugation still works because the tilt angle γ is invariant under z -rotations. Result:

$$V_s^{\text{tot}} = e^{-iN\delta_0\tau I_z} e^{-i\Omega_{\text{eff}}N\tau \hat{n}_s \cdot \vec{I}}, \quad \hat{n}_{0,1} = (\pm \cos \gamma, 0, \sin \gamma).$$

$$\frac{1}{2} \text{Tr}(U_0 U_1^\dagger) = 1 - \frac{2\Omega^2}{\Omega^2 + \delta_1^2} \sin^2 \left(\frac{\sqrt{\Omega^2 + \delta_1^2} N\tau}{2} \right)$$

The detuned-Rabi formula reproduces both the envelope and the side-lobe period:

DDrf: Suppression — Apodized Pulse Idea

Replace the constant RF amplitude with a per-cell envelope $\Omega_k = \Omega f(k)$, where f is a discrete window function (Hanning, Hamming, Blackman, ...):

Intuition. The side-lobes are essentially the discrete Fourier transform of a rectangular window. Shaping the window suppresses its sidelobes — the same trick used in classical signal processing.

The four windows we compare: rectangular (the baseline that produces the side-lobes), plus three classical apodization windows from signal processing.

Apodized envelopes flatten the off-resonant region while preserving the on-resonant peak. The effect grows with N (right): the rectangular window develops a clean side-lobe pair, while both apodized windows stay smooth.

DDrf: Suppression — Window Comparison

The normalized spectral response factorizes into a window-independent `sinc` and a window-dependent kernel:

$$\left| \frac{F(\delta_1)}{F(0)} \right|^2 = \text{sinc}^2(u) \cdot |G(u)|^2, \quad u = \frac{\delta_1}{2\Omega \bar{f}}.$$

For a rectangular (constant-amplitude) pulse,

$$\left| \frac{F}{F(0)} \right|_{\text{rect}}^2 = \text{sinc}^2(u).$$

For a Blackman-apodized pulse,

$$\left| \frac{F}{F(0)} \right|_{\text{Black}}^2 = \text{sinc}^2(u) \cdot \left(\frac{50u^4 - 209u^2 + 84}{21(u^2 - 1)(u^2 - 4)} \right)^2, \quad u = \frac{\delta_1}{2\Omega \bar{f}}.$$

The mismatch between consecutive axes is

$$\Delta\gamma_k \equiv \gamma_{k+1} - \gamma_k \approx -\frac{\delta_1}{\Omega} \Delta f_k,$$

where $\Delta f_k = f(k+1) - f(k)$. So the non-commutativity enters at order $\delta_1 \cdot \Delta f_k$, which is small when either δ_1 is small or the window varies slowly.

The full-evolution formalism above assumes Ω_{RF} is time-independent (compatible with the RWA). With a Gaussian envelope

$$\Omega_{\text{RF}}(t) = \Omega_0 e^{-(t-t_k)^2/2\sigma^2},$$

the per-segment propagator $e^{-iH'_{0,1}\tau}$ is no longer exact, and direct Schrödinger integration takes tens of minutes per frequency point.

Approach. Magnus expansion, valid when Ω_{RF} varies slowly relative to ω_{RF} .

For $\partial_t U = -iH(t)U$, write $U = e^{\Omega(t)}$ with $\Omega(t) = \sum_n \Omega_n(t)$:

$$\Omega_1 = \int_0^t A_1 dt_1, \quad \Omega_2 = \frac{1}{2} \iint [A_1, A_2] dt_1 dt_2, \dots$$

Computed terms (with $f(t)$ the Gaussian envelope):

$$\begin{aligned} \Omega_1(T) &= -i(\delta_{(0,1)} T I_z + c_1 (\cos \phi I_x + \sin \phi I_y)), \\ \Omega_3(T) &= \frac{\delta_{(0,1)}^2}{24} K_1 (\cos \phi I_x + \sin \phi I_y) + \frac{\delta_{(0,1)}}{24} K_2 I_z, \\ \Omega_2 &= \Omega_4 = 0, \end{aligned}$$

with $c_1 = \int_0^T f$ and $K_{1,2}$ triple integrals of f . Convergence is guaranteed when $\int_0^T \|A(s)\|_2 ds < \pi$ — likely satisfied here; **simulation pending**.

Special thanks to WD Lee.

DONE

Mattermost: private channel

WIP

Send alert from Outline changes to Mattermost

MAYBE

Code / PPT update

WIP

Cool-paper channel update to Outline

MAYBE

Equipment directory / Diamond sample

DONE

Emergency contact list

An alert is sent when:

- a new document is added, a comment is added, or a document is modified;
- and only if the *collection* has a corresponding Mattermost channel.

- Notion-like dataview (requires SQL-like embedding).
- Table, Kanban, Calendar, and timeline views are supported.

Datafield values are not searchable as keywords on their own — the dataview feature on the previous slide is the workaround.

- Set up a desktop with an RTX 3090, accessible remotely from anywhere.
- Borrowing a server from Prof. Yosep Kim (Korea University).
- (personal) Promoted to principal candidate for the Fulbright STEM scholarship on Apr 16; Cornell Applied Physics initially responded positively but ultimately rejected my application, so I formally declined the Fulbright STEM scholarship.
- (personal) Now an official research intern here!

Thank You!

Questions & Discussion